Foundations of Large-Scale Multimedia Information Management and Retrieval

Lecture #4 Similarity

Edward Y. Chang

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Foundations of Large-Scale Multimedia Information Management and Retrieval
Mathematics of Perception

Foundations of Large-Scale Multimedia Information Management and Retrieval *Mathematics of Perception* covers knowledge representation and semantic analysis of multimedia data and scalability in signal extraction, data mining, and indexing. The book is divided into two parts: Part I - Knowledge Representation and Semantic Analysis focuses on the key components of mathematics of perception as it applies to data management and retrieval. These include feature selection/reduction, knowledge representation, semantic analysis, distance function formulation for measuring similarity, and multimodal fusion. Part II - Scalability Issues presents indexing and distributed methods for scaling up these components for high-dimensional data and Web-scale datasets. The book presents some real-world applications and remarks on future research and development directions.

The book is designed for researchers, graduate students, and practitioners in the fields of Computer Vision, Machine Learning, Large-scale Data Mining, Database, and Multimedia Information Retrieval.

Dr. Edward Y. Chang was a professor at the Department of Electrical & Computer Engineering, University of California at Santa Barbara, before he joined Google as a research director in 2006. Dr. Edward Y. Chang received his M.S. degree in Computer Science and Ph.D degree in Electrical Engineering, both from Stanford University.

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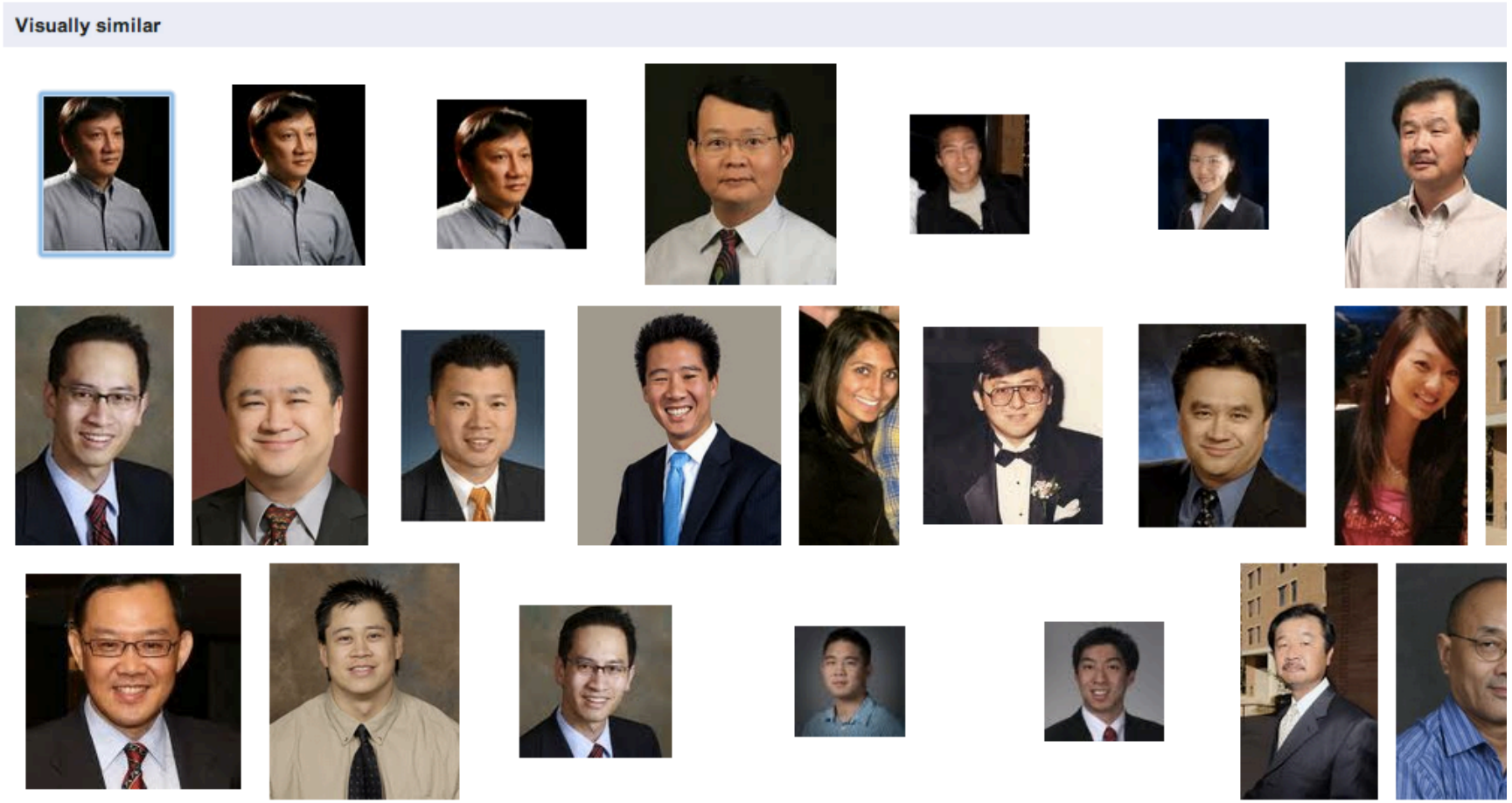
Foundations of Large-Scale Multimedia Information Management and Retrieval

Mathematics of Perception

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Similar?



Two Key Technical Problems

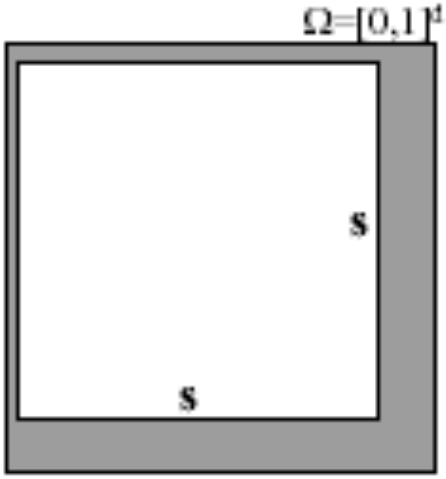
- Curse of Dimensionality
- Modeling Subjectivity
  - Query/User/App Dependent

Dimensionality Curse

- D: Data Dimension
- When D increases
  - Nearest neighbors are not local
  - All points are equally distanced

Sparse High-D Space [C. Aggarwal, etc. ICDT 2001]

- Hyper-cube Range Queries



$$P^d[s] = s^d$$

Range Coverage → 0%

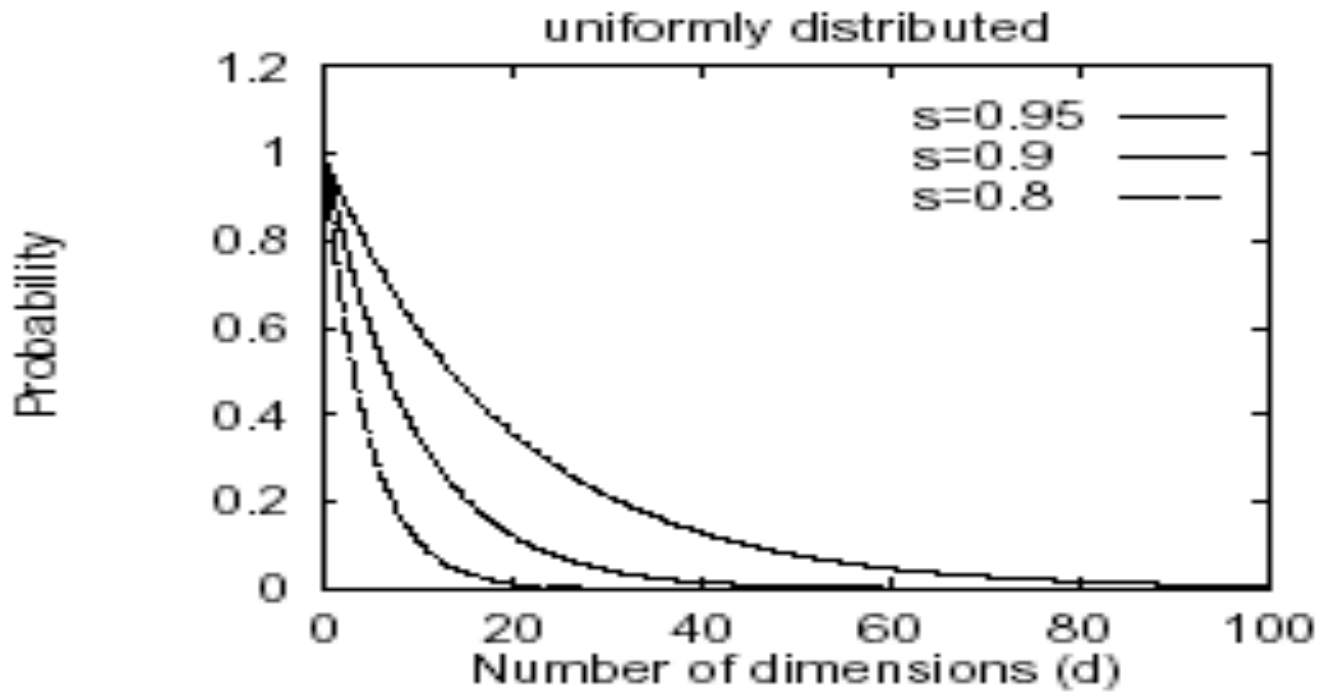
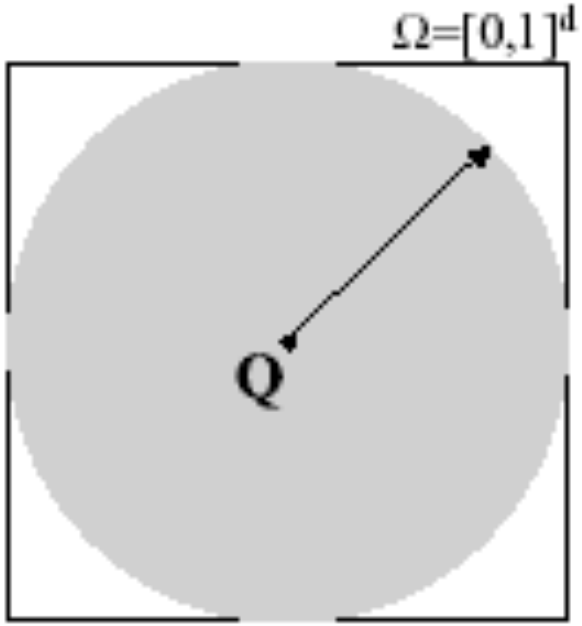


Figure 2: The probability function $P^d[s]$.

Sparse High-D Space

- Spherical Range Queries



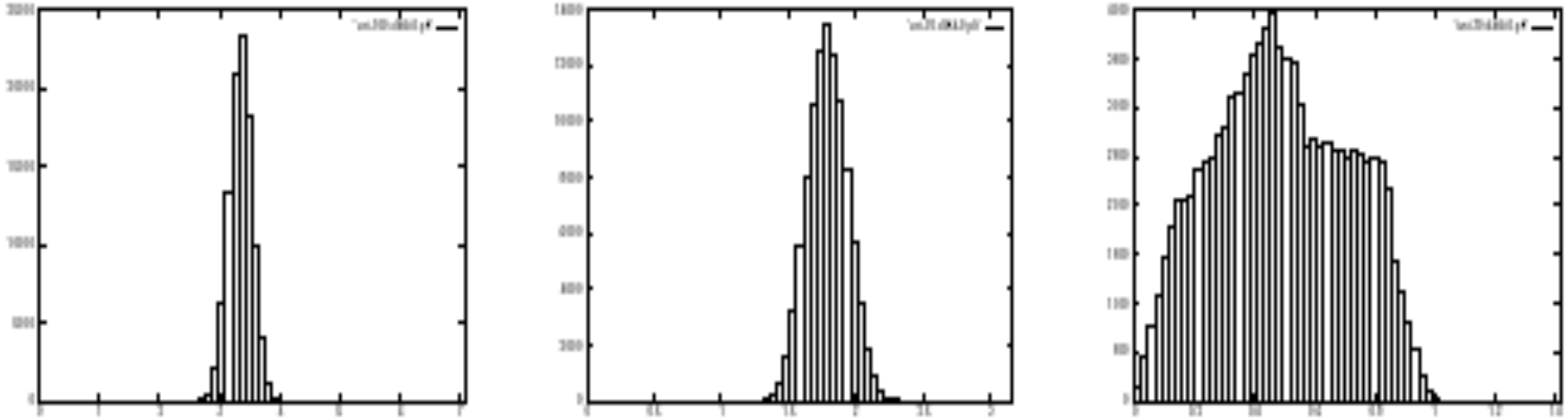
$$P[R \in sp^d(Q, 0.5)] = \frac{\sqrt{\pi^d} \bullet (0.5)^d}{\Gamma(\frac{d}{2} + 1)}$$

No Point in the Nearest Neighborhood

d	$P[R \in sp^d(Q, 0.5)]$	$N(d)$
2	0.785	1.273
4	0.308	3.242
10	0.002	401.5
20	$2.461 \cdot 10^{-8}$	40'631'627
40	$3.278 \cdot 10^{-21}$	$3.050 \cdot 10^{20}$
100	$1.868 \cdot 10^{-70}$	$5.353 \cdot 10^{69}$

Table 2: Probability that a point lies within the largest range query inside Ω, and the expected database size.

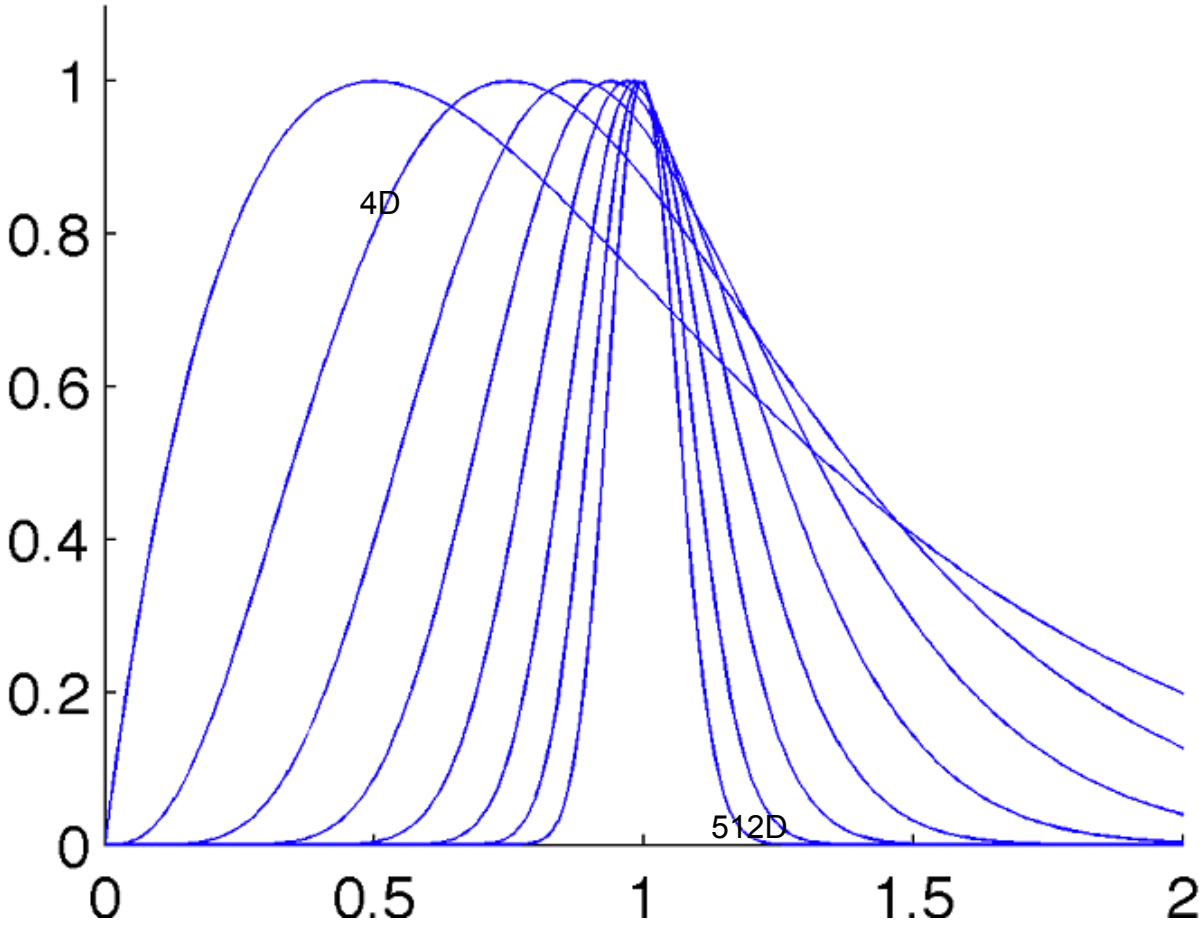
Dimensionality Curse



(a) 50 Dimensions (b) 10 Dimensions (c) 2 Dimensions

Figure 3: Distance Distribution of Uniform Data

Equidistant Points



Are We Doomed?

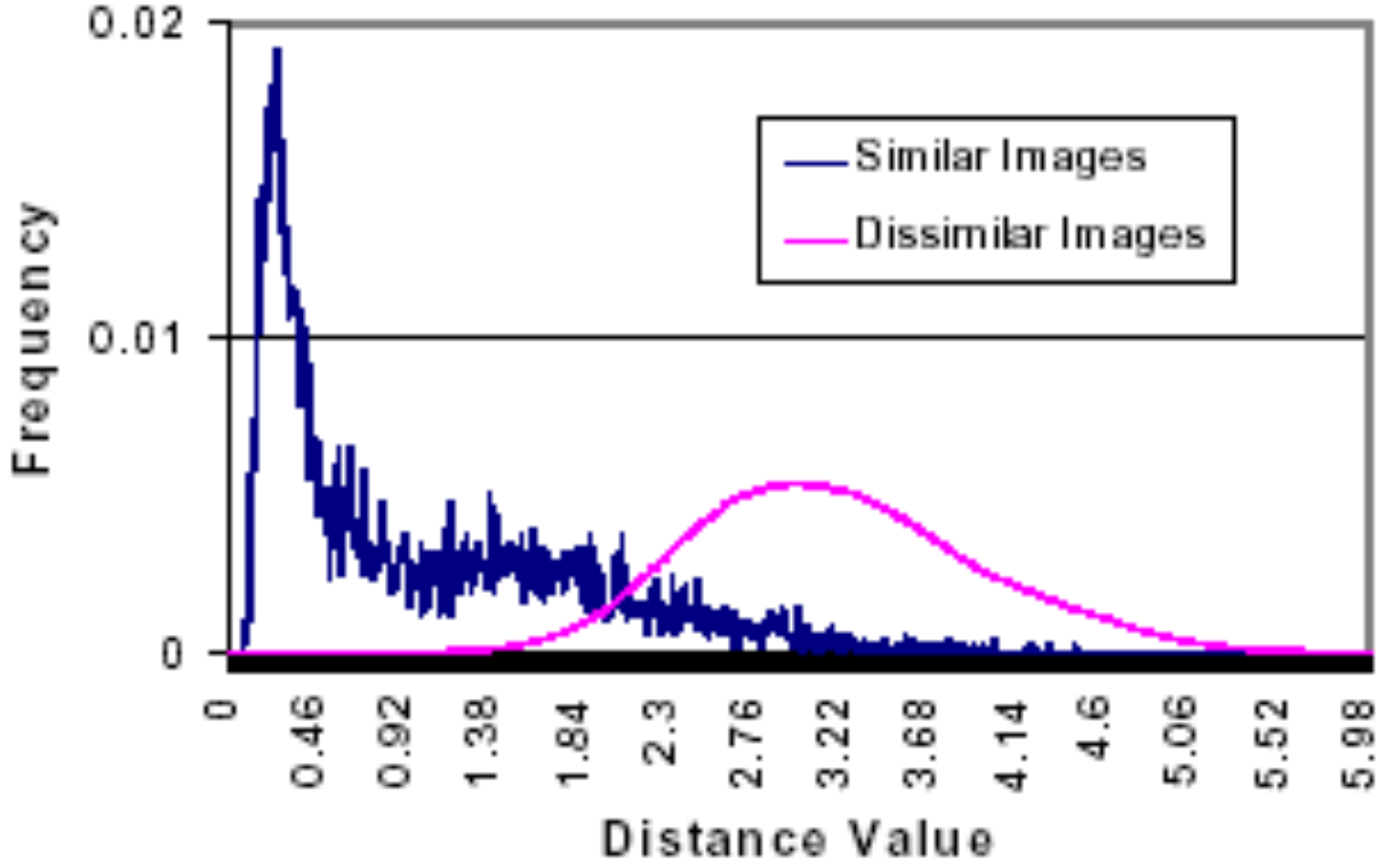
- How does the curse affect classification?

- Similar objects tend to cluster together
- Dimensionality reduction

Summary of Approaches

- Dynamic Partial Function
- Restricted Estimators
  - Specifying the nature of local neighborhood
  - E.g., Manifold learning
- Adaptive Feature Reduction
  - PCA, LDA

Distribution of Distances



(a) $m = 144$

Some Solutions to High-D

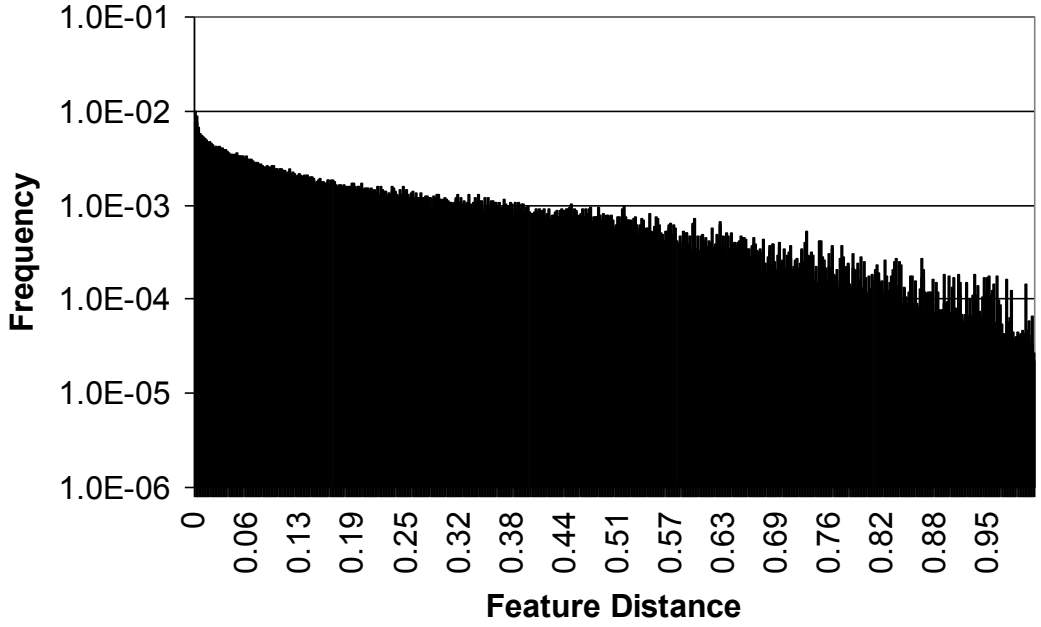
- Restricted Estimators
  - Specifying the nature of local neighborhood
  - Manifold learning
- Adaptive Feature Reduction
  - PCA, LDA
- Dynamic Partial Function

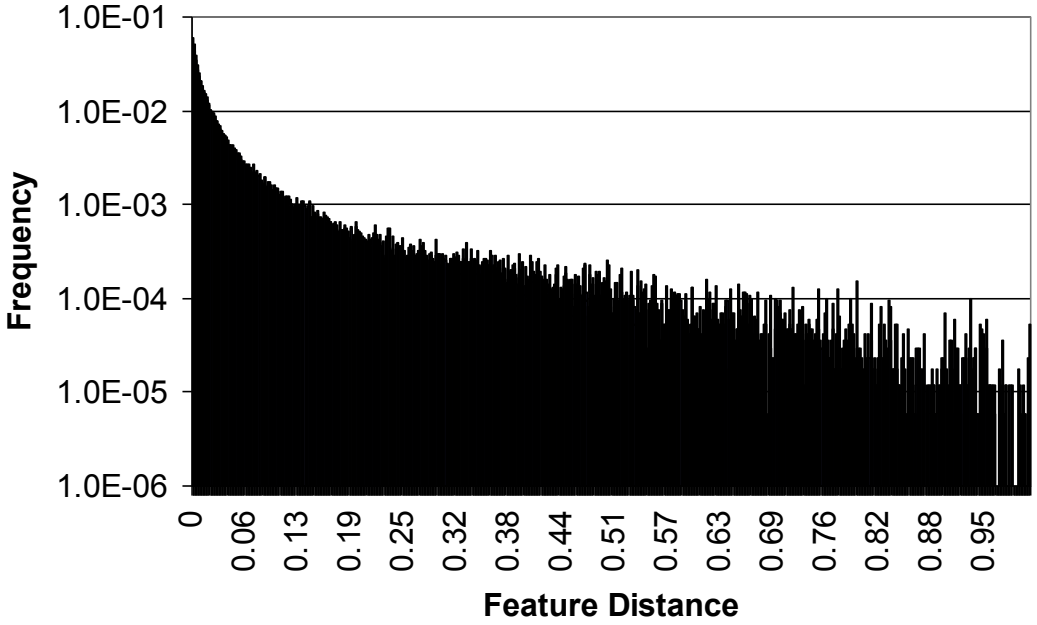
Three Major Paradigms

- Preserve data description in a lower dimensional space
  - PCA
- Maximize discriminability in a lower dimensional space
  - LDA
- Activate only similar channels
  - DPF

Minkowski Distance

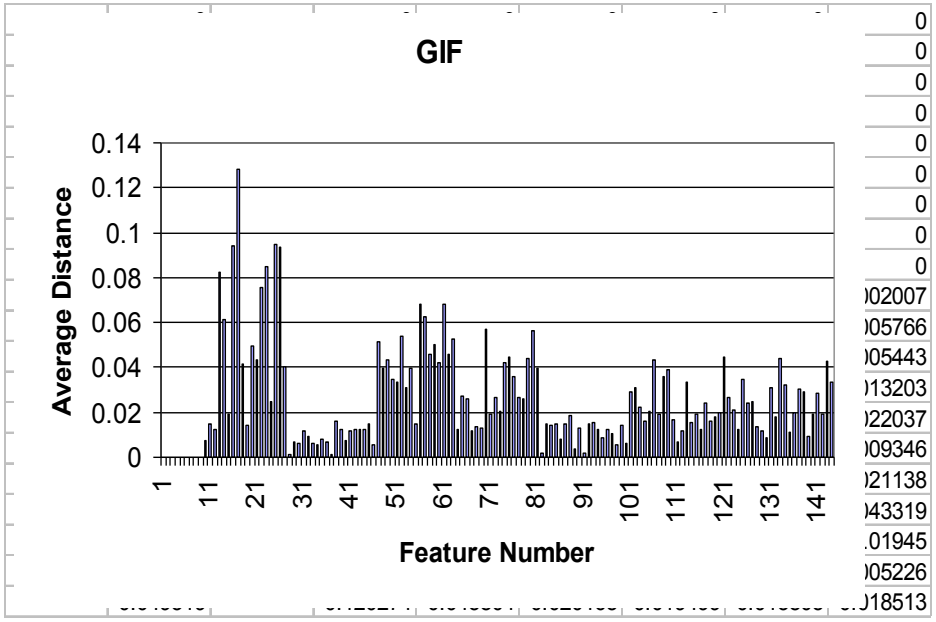
- Objects P and Q
- $D = (\Sigma_M (p_i - q_i)^n)^{1/n}$
- Similar images are similar in all M features

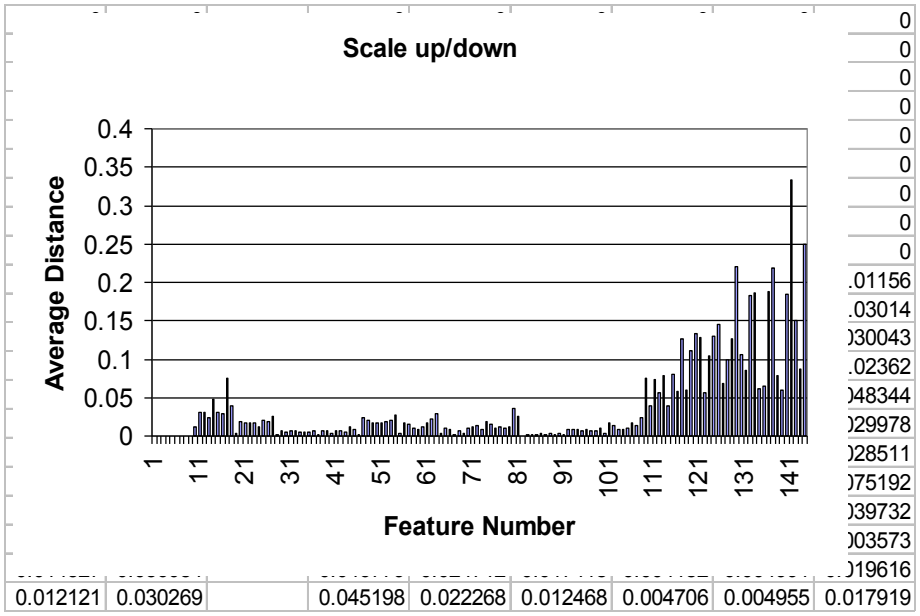

1.0E-01
1.0E-02
1.0E-03
1.0E-04
1.0E-05
1.0E-06
Frequency
0
0.06
0.13
0.19
0.25
0.32
0.38
0.44
0.51
0.57
0.63
0.69
0.76
0.82
0.88
0.95
Feature Distance

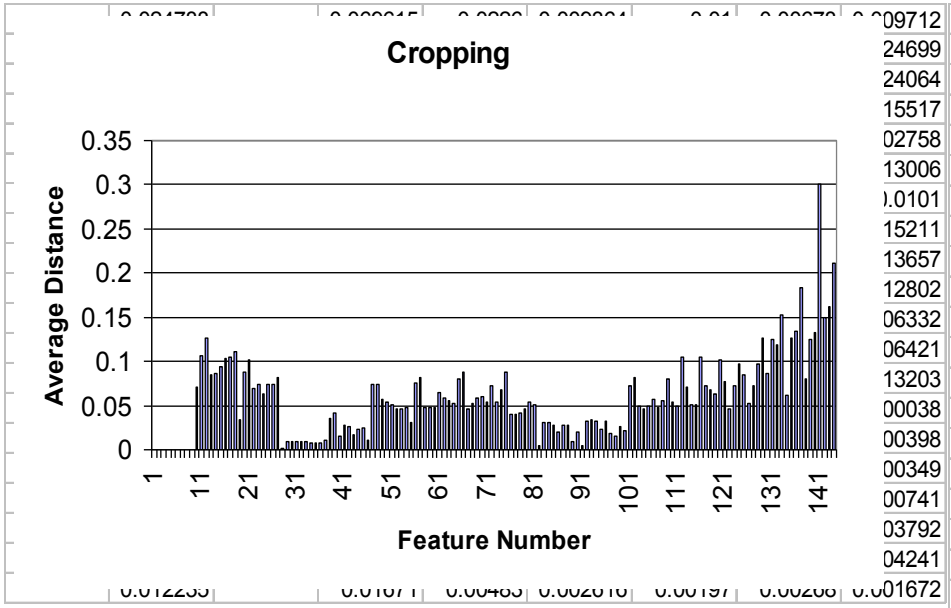

1.0E-01
1.0E-02
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1.0E-04
1.0E-05
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0.88
0.95
Feature Distance

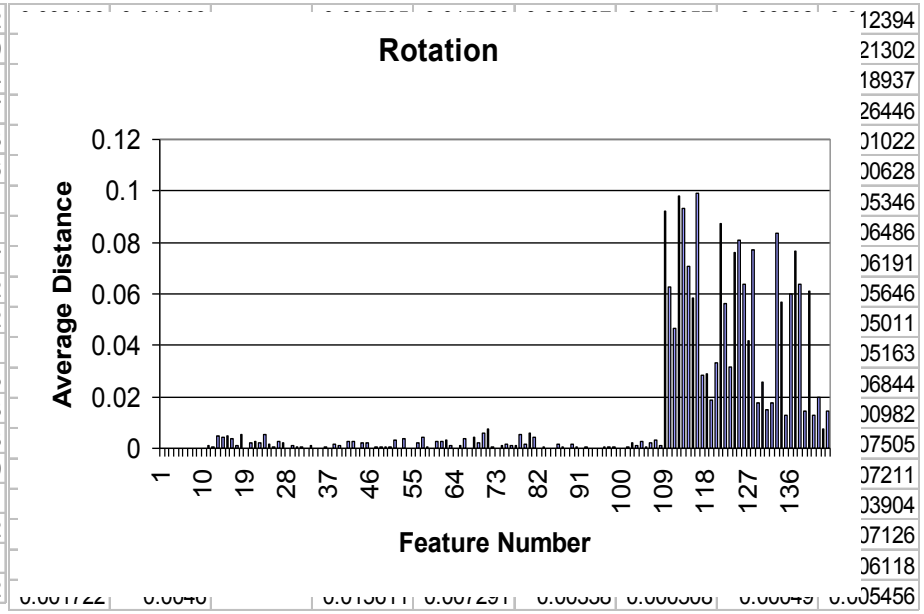
Weighted Minkowski Distance

- $D = (\Sigma_M w_i(p_i - q_i)^n)^{1/n}$
- Similar images are similar in the same subset of the M features


GIF
Average Distance
Feature Number
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
1
11
21
31
41
51
61
71
81
91
101
111
121
131
141


Scale up/down
Average Distance
Feature Number
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
1
11
21
31
41
51
61
71
81
91
101
111
121
131
141


Cropping
Average Distance
Feature Number
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
1
11
21
31
41
51
61
71
81
91
101
111
121
131
141


Rotation
Average Distance
Feature Number
0.12
0.1
0.08
0.06
0.04
0.02
0
1
10
19
28
37
46
55
64
73
82
91
100
109
118
127
136

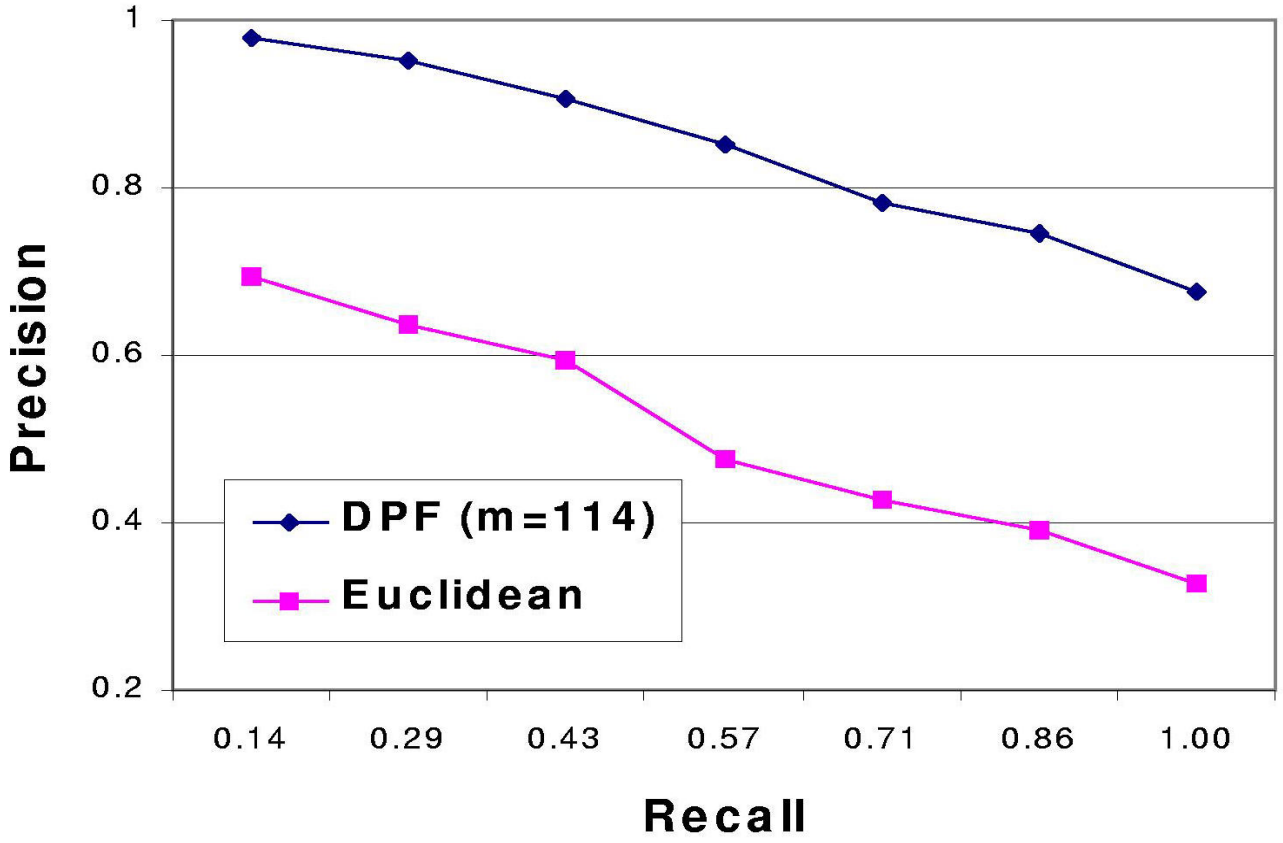
Similarity Theories

- Objects are similar in all respects (Richardson 1928)
- Objects are similar in some respects (Tversky 1977)
- Similarity is a process of determining respects, rather than using predefined respects (Goldstone 94)

DPF

- Which Place is Similar to Kyoto?
- Partial
- Dynamic
- Dynamic Partial Function

Precision/Recall



Summary of Approaches

- Dynamic Partial Function
- Restricted Estimators
  - Specifying the nature of local neighborhood
  - E.g., Manifold learning
- Adaptive Feature Reduction
  - PCA, LDA

Manifold Learning Algorithms

	Auto. NN	KPCA	Principal curves	SOM	GTM	MDS	ISOMAP	LLE
Explicit Manifold	No	No	Yes	No	Yes	No	Yes	Yes
Parametric	Yes	Yes	No	No	Yes	No	No	No
Dissimilarity matrix	No	No(?)	No	No	No	Yes	Yes	No(?)
Local neighborhood	No	No	No(?)	No	No	No	Yes	Yes

Geodesic Distance

- Geodesic: the shortest curve on a manifold that connects two points on the manifold
  - Example: on a sphere, geodesics are great circles
- Geodesic distance: length of the geodesic

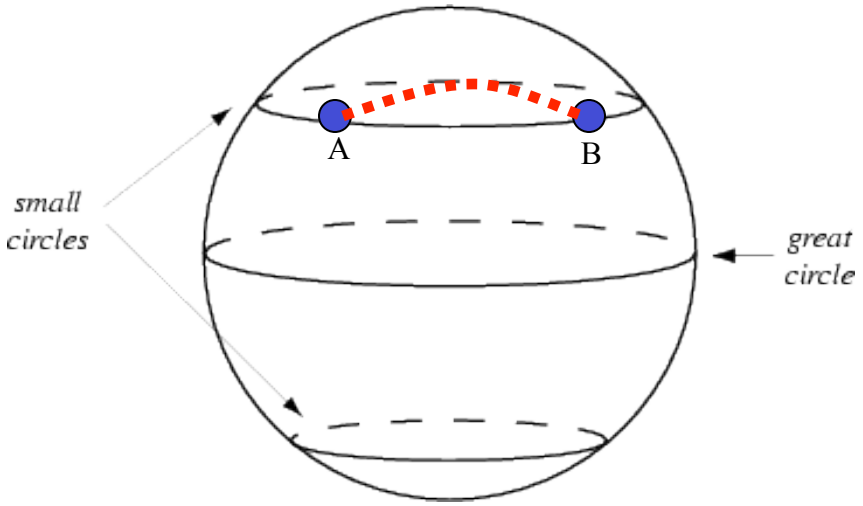


Figure from http://mathworld.wolfram.com/GreatCircle.html

Geodesic Distance

- Euclidean distance needs not be a good measure between two points on a manifold
  - Length of geodesic is more appropriate
- Example: Swiss roll

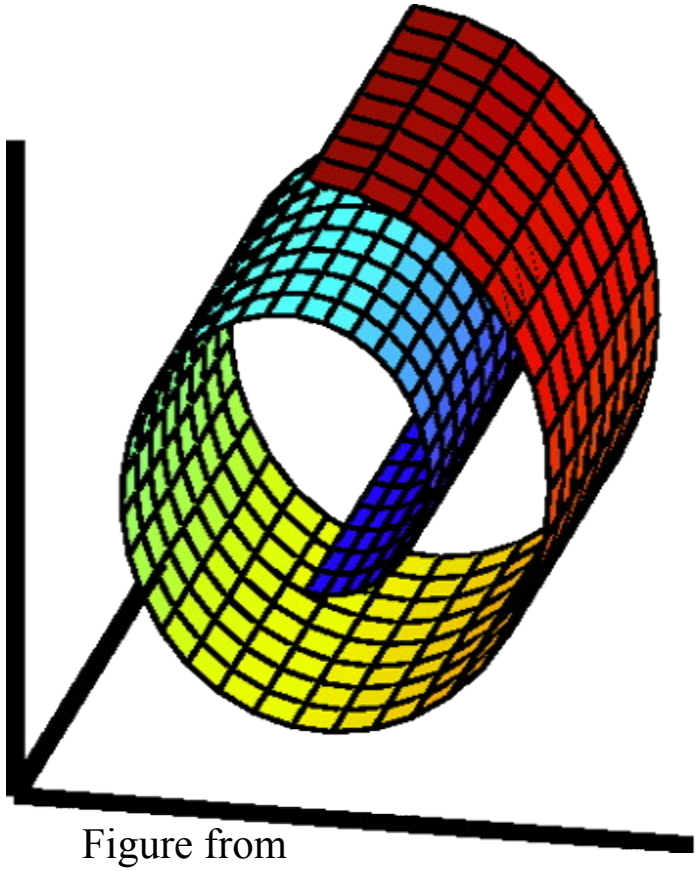


Figure from LLE paper

Isometric Feature Mapping (ISOMAP)

- Take a distance matrix $\{g_{ij}\}$ as input
- Estimate geodesic distance between any two points by "a chain of short paths"
  - Formulate this as a graph theory problem
- Perform classical scaling on the matrix of geodesic distances to obtain final projection

Steps to Estimate Geodesic Distances

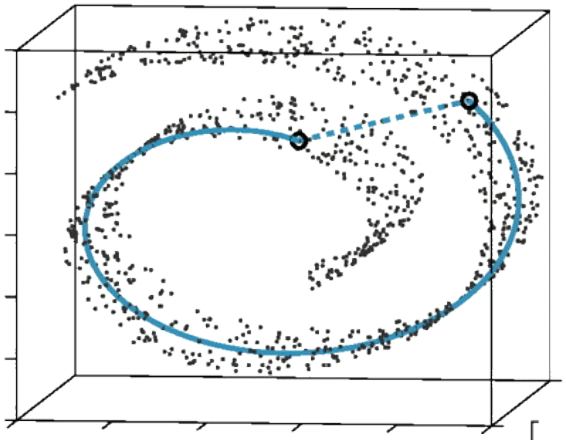
1. Find the "neighbors" of all data items $\mathbf{z}_i$
   - Two possible definitions of neighbors
     - Set of items whose distances are less than e
     - The *K* closest items
2. Construct a weighted undirected graph
   - Vertex *i* corresponds to $\mathbf{z}_i$
   - An edge between the vertex *i* and *j* iff $\mathbf{z}_i$ and $\mathbf{z}_j$ are neighbors, and its weight is g_{ij}

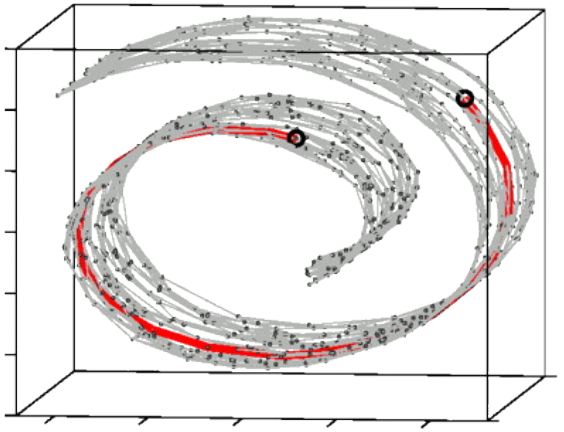
Steps to Estimate Geodesic Distances

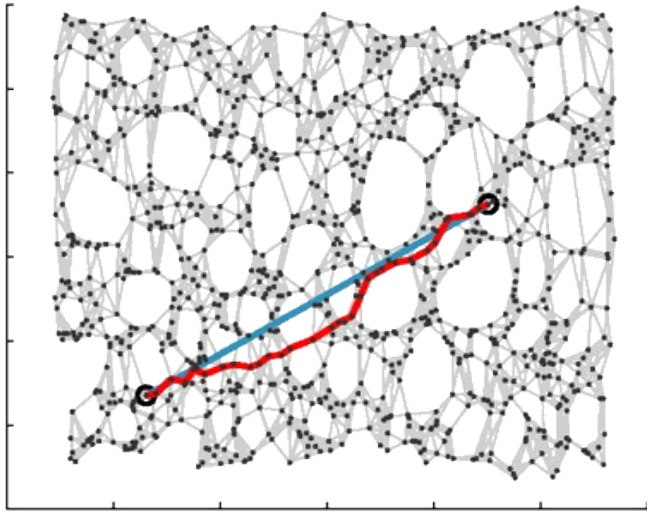
3. Find the shortest distance between all pairs of vertices in the graph
   - Floyd ($O(m^3)$) or Dijkstra ($O(m^2 \log m + mp)$)

- The shortest distance between vertices *i* and *j* in the graph is the estimated geodesic distance between $\mathbf{z}_i$ and $\mathbf{z}_j$

Rationale for the Geodesic Distance Estimation







Figures from ISOMAP paper

A Run of ISOMAP

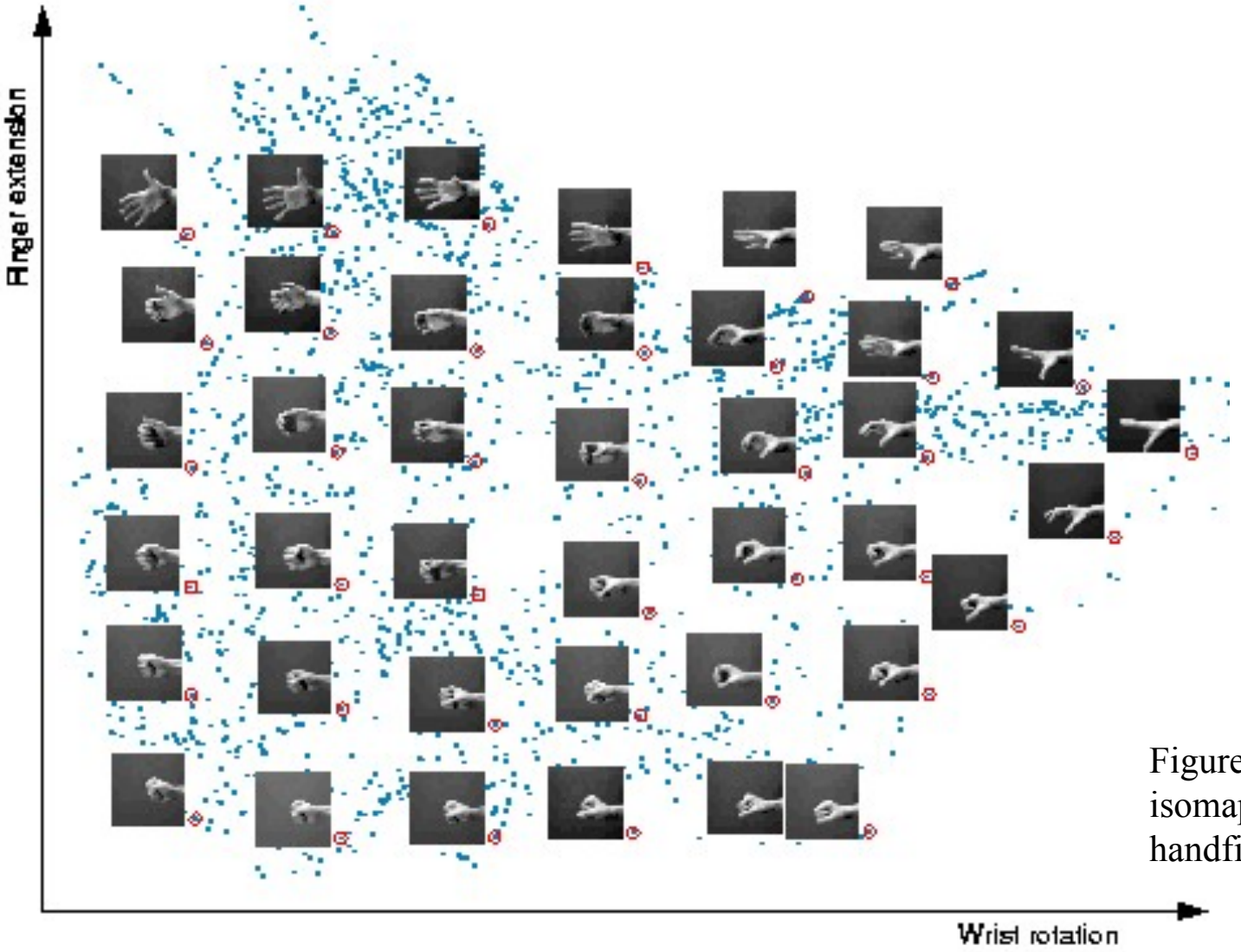
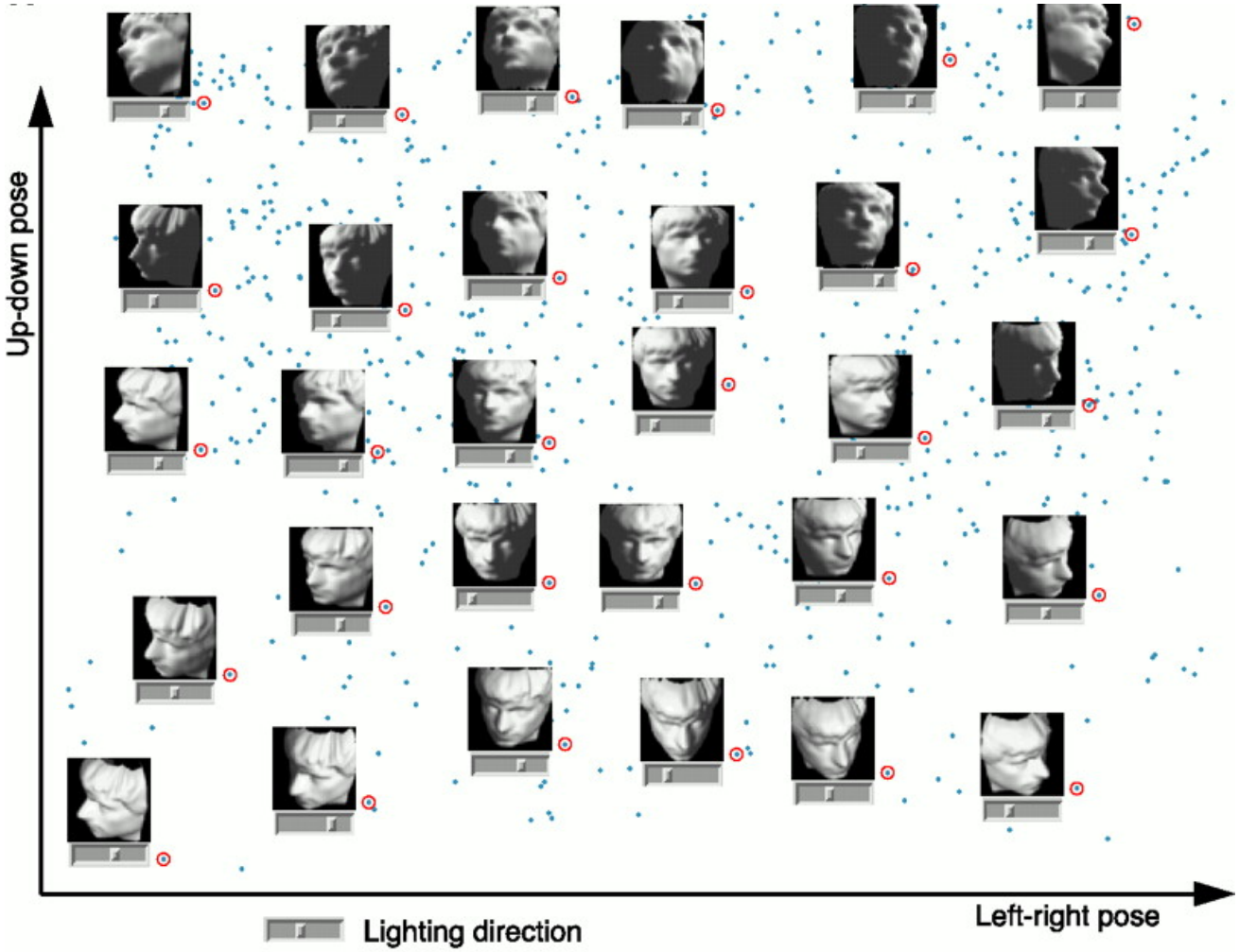


Figure from http://isomap.stanford.edu/handfig.html

A Run of ISOMAP



Figures from ISOMAP paper

Interpolation on Straight Lines in the Projected Co-ordinates



Figures from ISOMAP paper

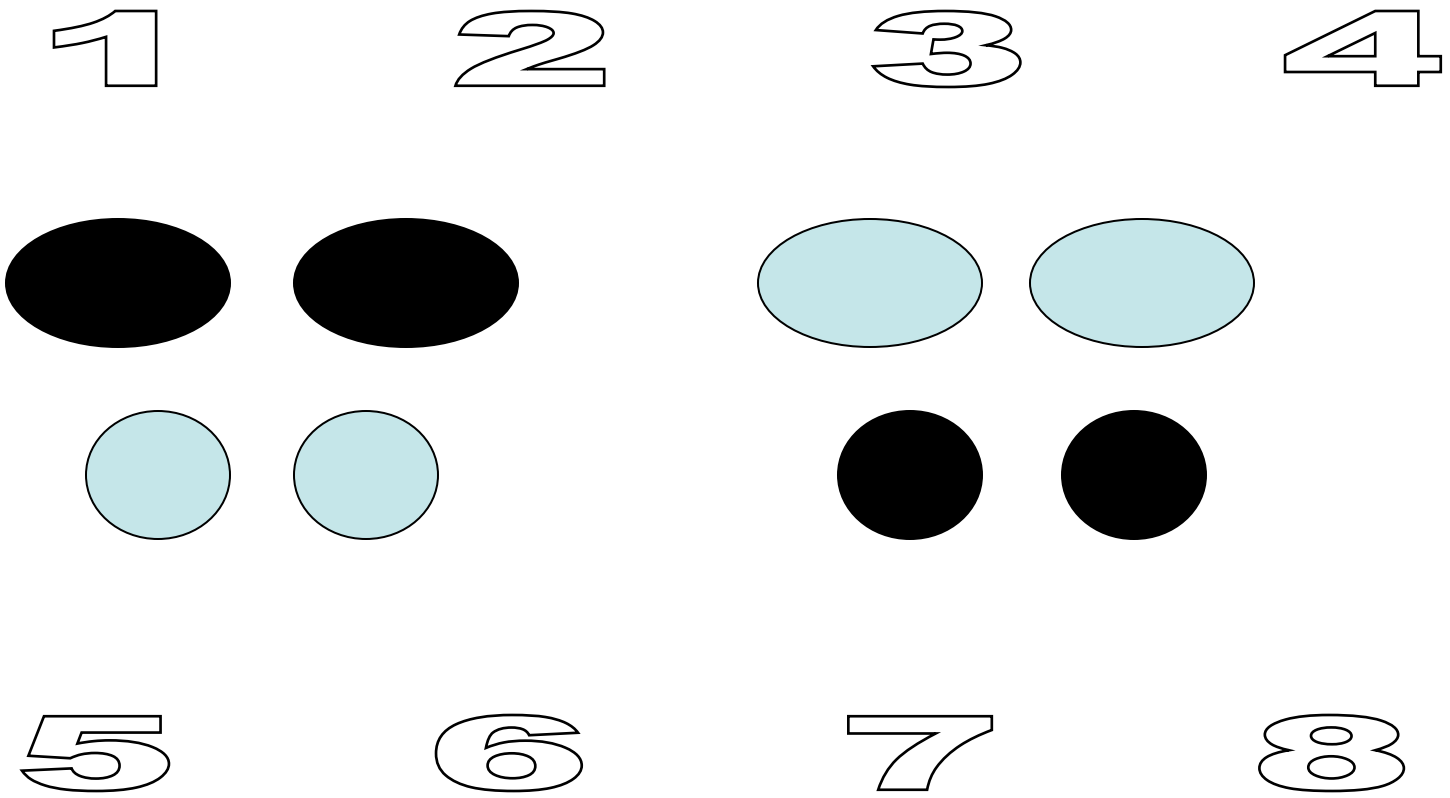
Summary of Approaches

- Dynamic Partial Function
- Restricted Estimators
  - Specifying the nature of local neighborhood
  - E.g., Manifold learning
- Adaptive Feature Reduction
  - PCA, LDA

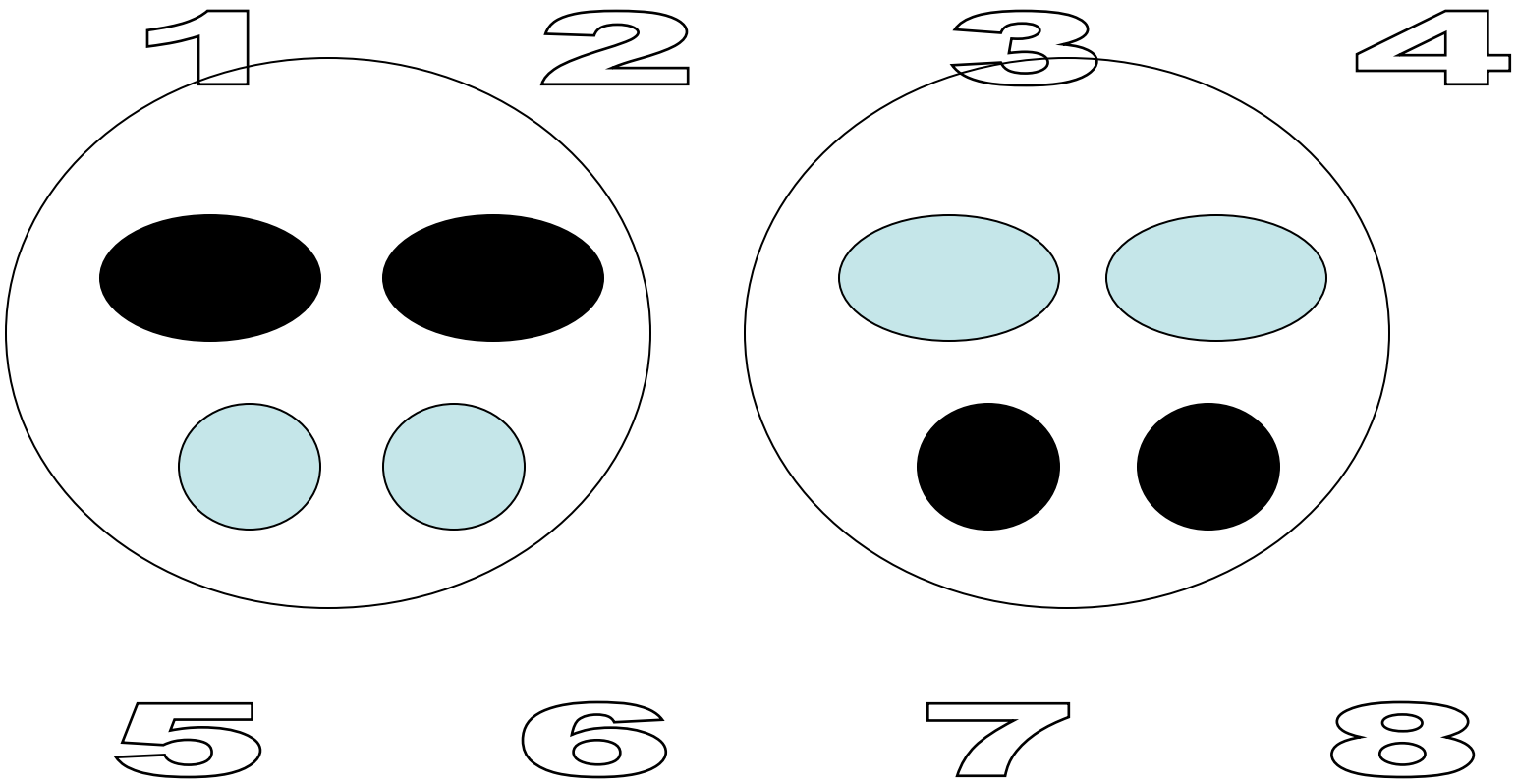
Two Key Technical Problems

- Curse of Dimensionality
- Modeling Subjectivity
  - Query/User/App Dependent

Distance Function?



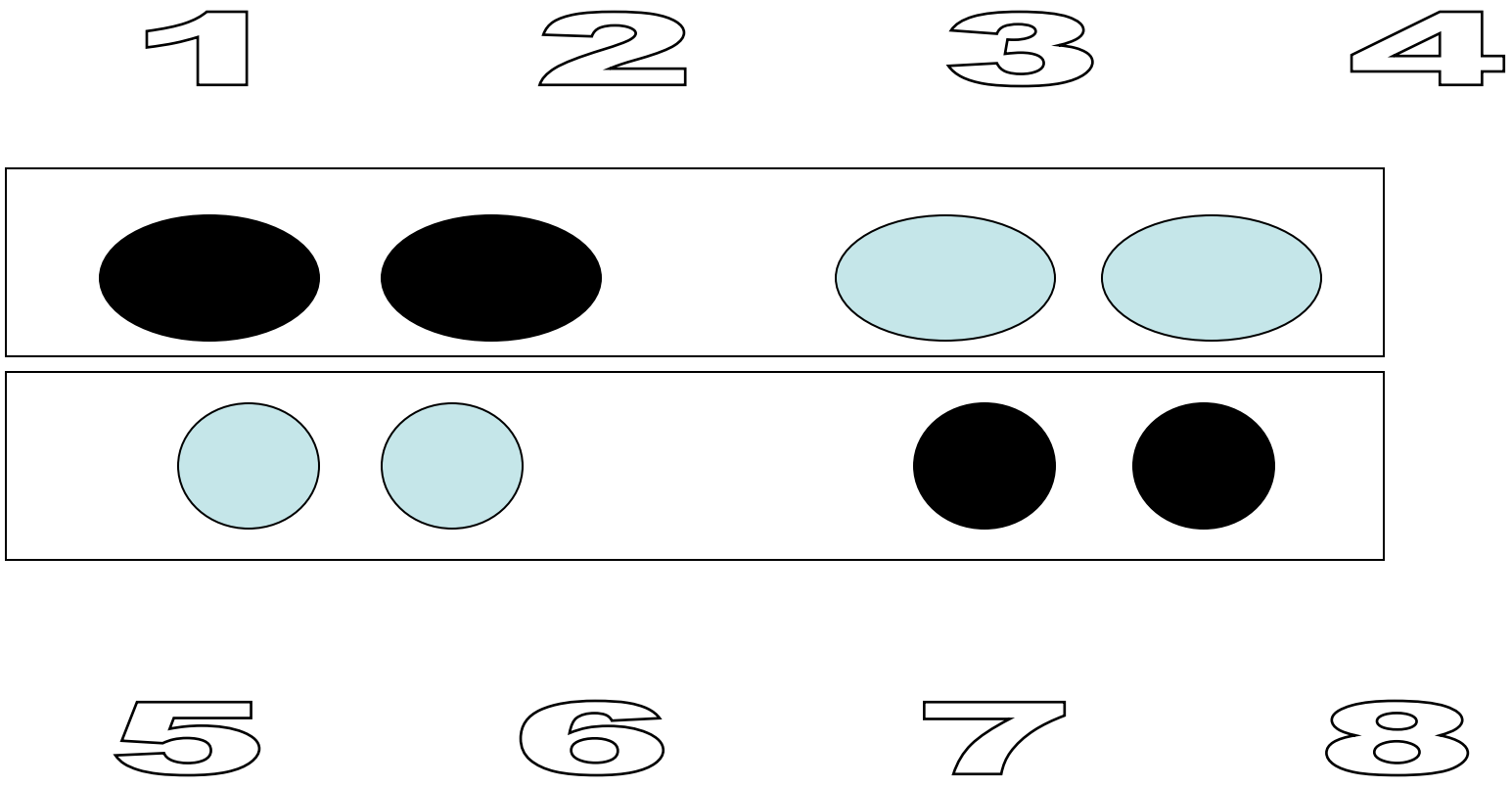
Group by Proximity



Group by Proximity

	x1	x2	x3	x4	x5	x6	x7	x8
x1	1	.7	.4	.3	.7	.6	.2	.1
X2		1	.4	.3	.6	.7	.3	.2
X3			1	.7	.3	.4	.7	.6
x4				1	.1	.2	.6	.7
x5					1	.7	.3	.2
x6						1	.6	.4
x7							1	.7
X8								1

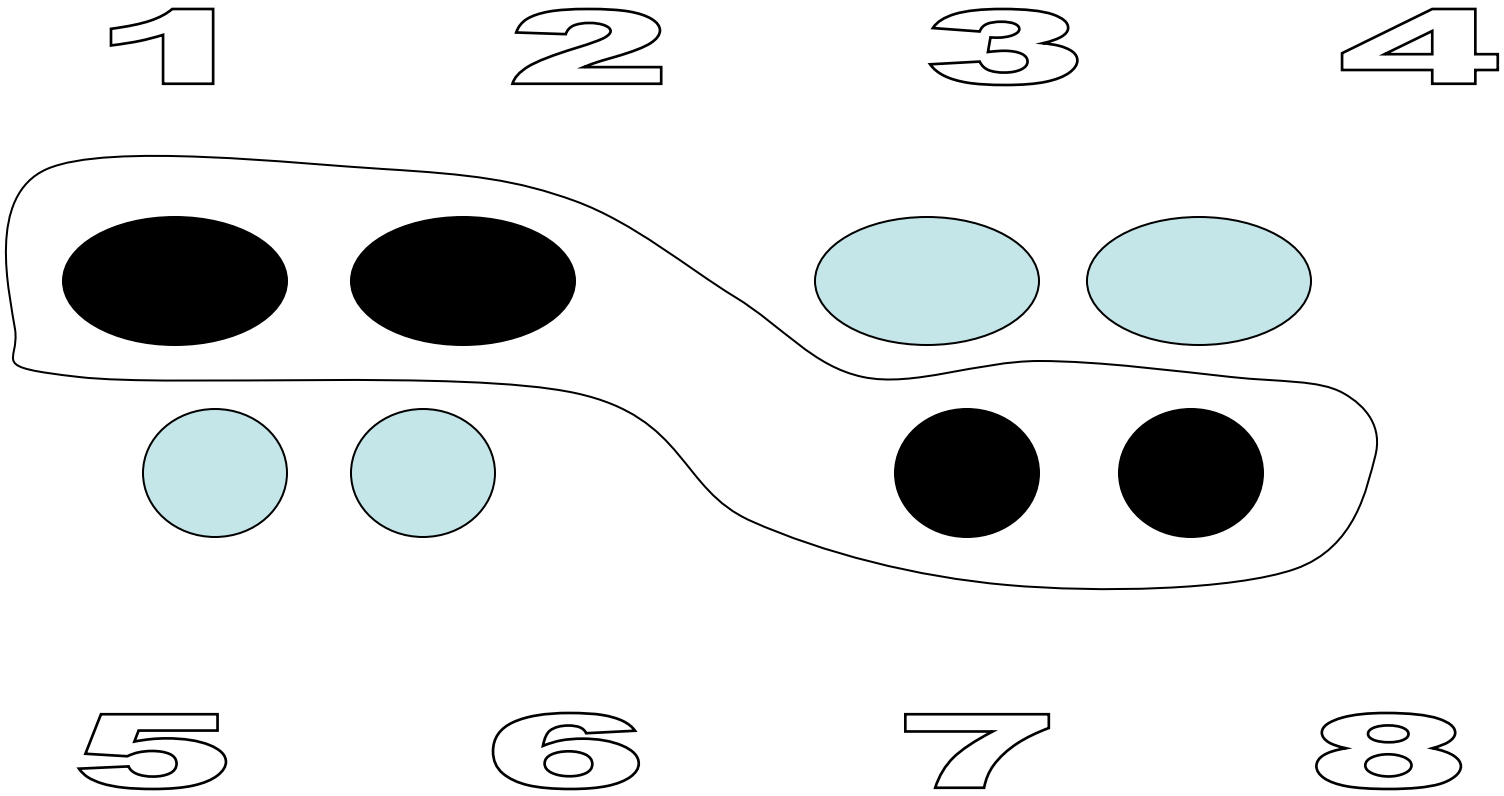
Group by Shape



Group by Shape

	x1	x2	x3	x4	x5	x6	x7	x8
x1	1	.7	.7	.7	.2	.2	.2	.2
X2		1	.7	.7	.2	.2	.2	.2
X3			1	.7	.2	.2	.2	.2
x4				1	.2	.2	.2	.2
x5					1	.7	.7	.7
x6						1	.7	.7
x7							1	.7
X8								1

Group by Color



Group by Color

	x1	x2	x3	x4	x5	x6	x7	x8
x1	1	.7	.3	.3	.3	.2	.2	.7
x2		1	.3	.3	.3	.3	.7	.7
x3			1	.7	.7	.7	.3	.3
x4				1	.7	.7	.3	.3
x5					1	.7	.3	.3
x6						1	.3	.3
x7							1	.7
x8								1

Naïve Alignment Rules

- Increasing the scores of similar pairs
- Decreasing the scores of dissimilar pairs
- $S_{ij} > D_{ij}$

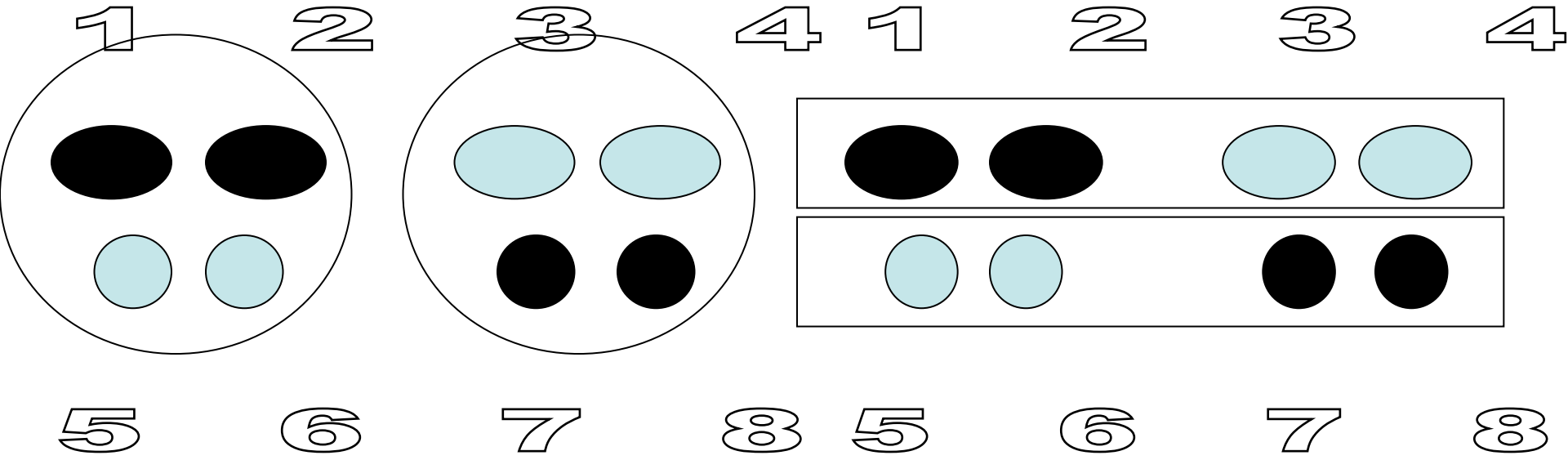
Our Work [ACM KDD 2005, ACM MM 05]

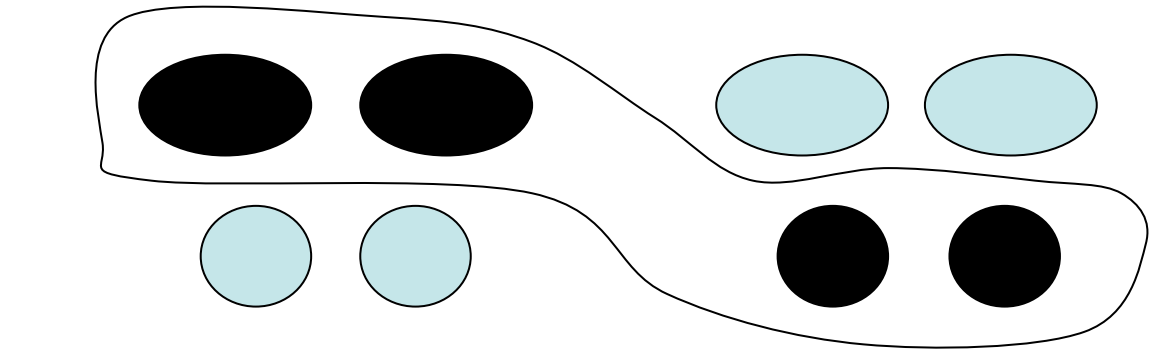
- kij = β_1 kij if (xi, xj) $\in$ D
- kij = β_2 kij + (1 - β_2) if (xi, xj) $\in$ S
- $0 \leq \beta_1 \beta_2 \leq 1$

- Theorem #1 The resulting matrix is psd
- Theorem #2 The resulting matrix is better aligned with the ideal kernel

Personalization & Scalability

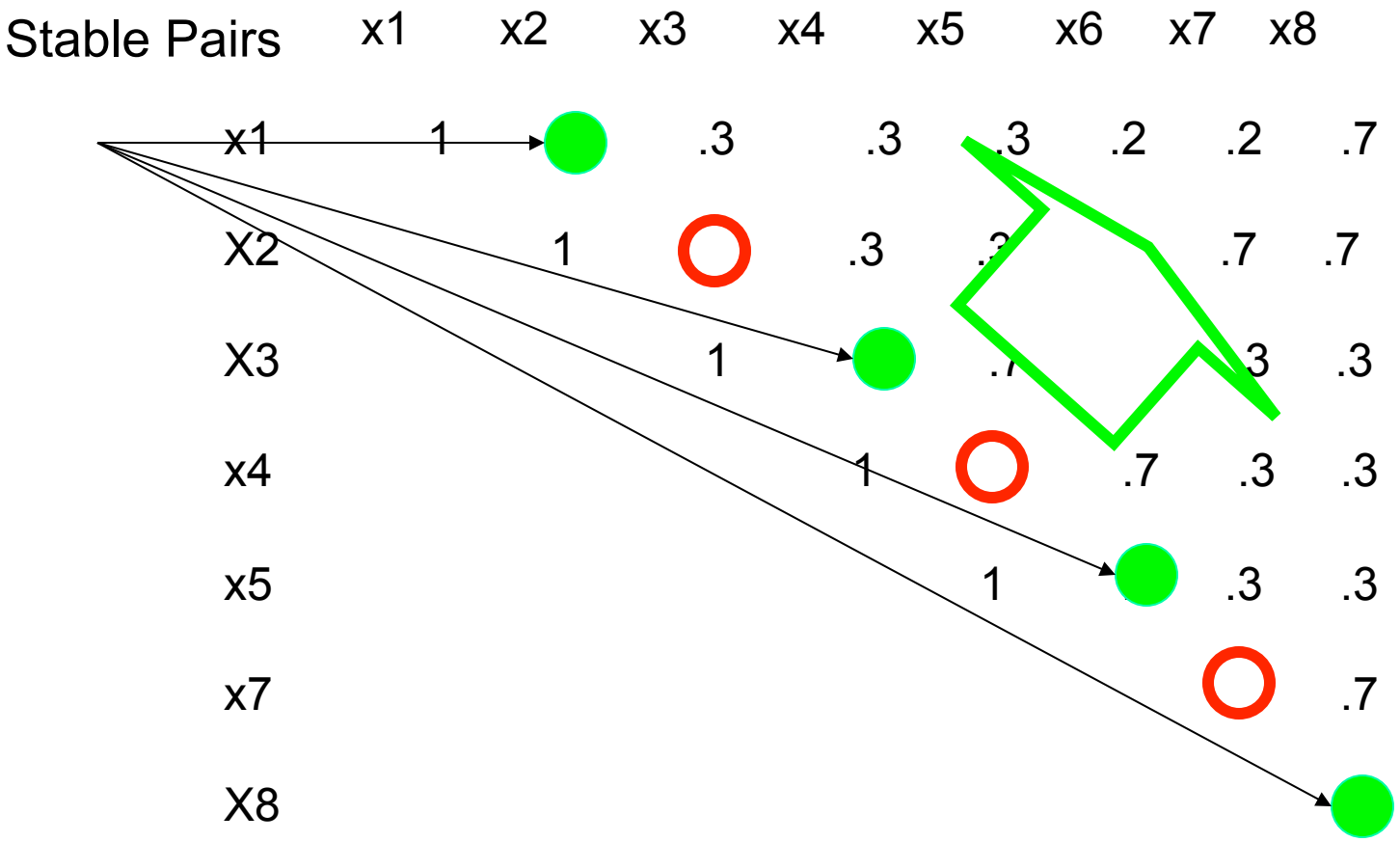
- Unsupervised Method
  - Clustering
  - Multi-version Clustering
- Active Learning
- Reinforcement Learning





Pairs 1,2
3,4
5,6
7,8
Are stable pairs

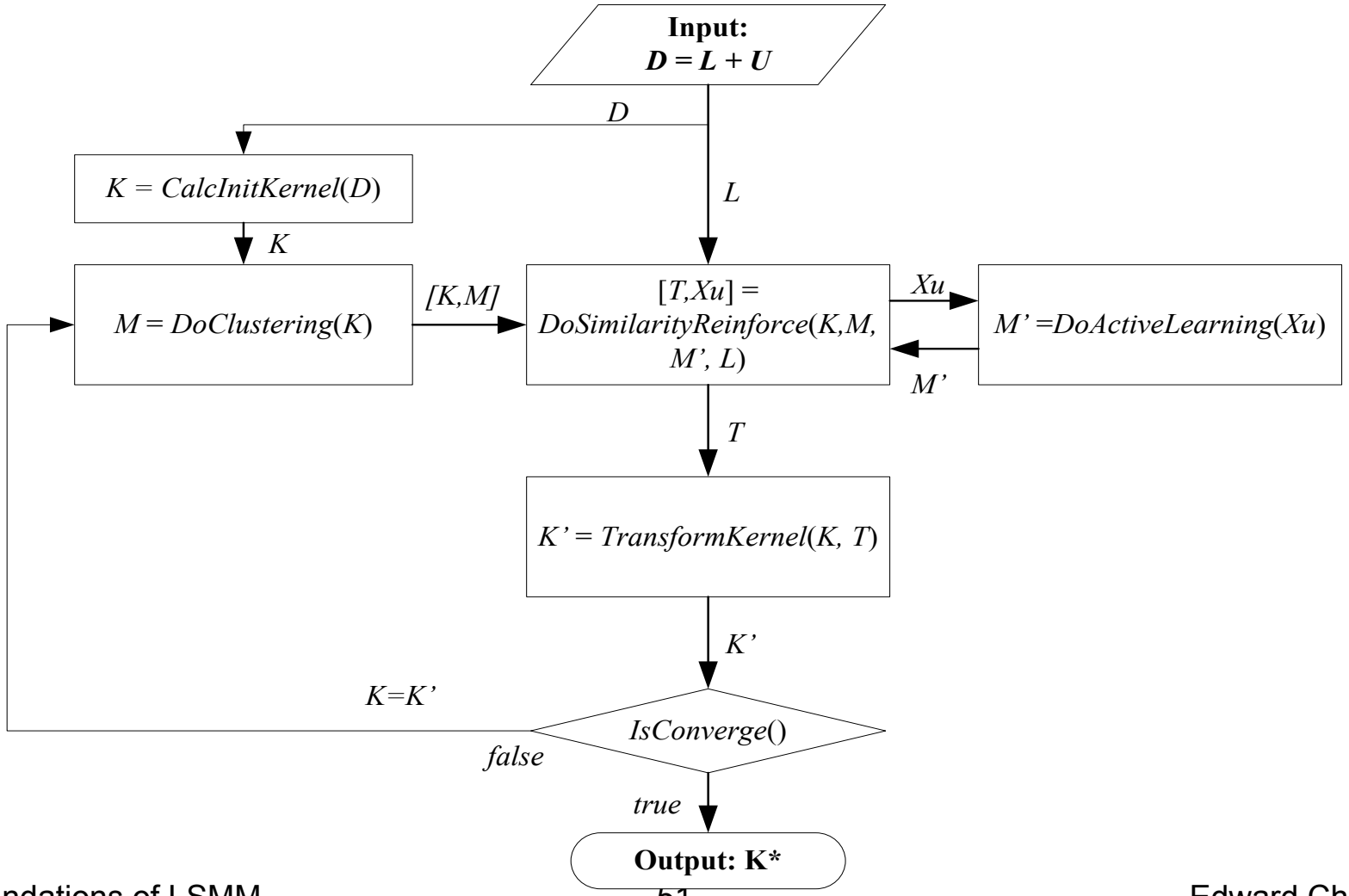
ULP: Unified Learning Paradigm



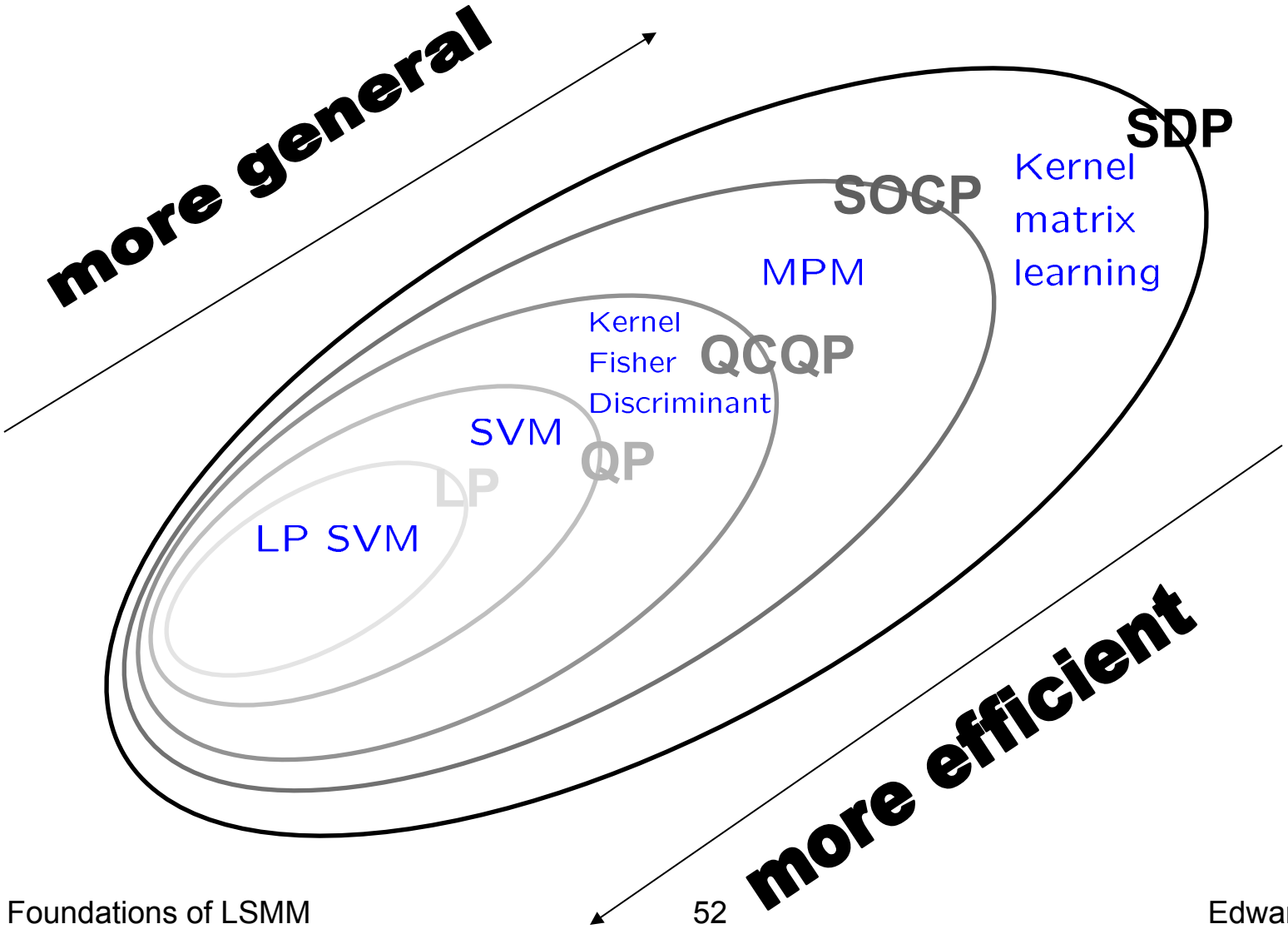
ULP

- Stable Pairs (green circles)
  - Found via shot-gun clustering
- Selected Uncertain Pairs (red circles)
  - Identified via the maximum information or fastest convergence rule
- Propagation (green arrow)

ULP [EITC 05]


Input:
D = L + U
D
L
K = CalcInitKernel(D)
K
M = DoClustering(K)
[K,M]
[T,Xu] = DoSimilarityReinforce(K,M, M', L)
Xu
M' =DoActiveLearning(Xu)
M'
T
K' = TransformKernel(K, T)
K'
K=K'
IsConverge()
false
true
Output: K*

Convex Optimization



Learning Similarity from Data

- Please refer to Chap 5 of FLSMIMR

Summary

- Curse of Dimension
  - Dynamic Partial Function
  - Manifold learning
  - PCA, LDA
- Learning Distance Function from Data
  - Kernel Alignment
  - Unified Learning Paradigm

Reading

- Foundations of Large-Scale Multimedia Information Management and Retrieval, E. Y. Chang, Springer, 2011
  - Chapter #4 Similarity
  - Chapter #5 Learning Distance Function